# 問題 1

I. 十分に広い 2 枚の導体板が真空中に設置されている．図 1 に示すように，導体板 A，B は $x = 0$ および $x = H(> 0)$ の位置に平行に設置され，それぞれ $\sigma_{\mathrm{A}}$ および $\sigma_{\mathrm{B}}$ の一様な面電荷密度で帯電している．導体板の厚さおよび端効果は無視でき，各導体板上の電荷がつくる電界は $x$ 方向成分のみを持つものとする．真空の誘電率を $\varepsilon_0$ とし，以下の問に答えよ．

(1) 各導体板上の電荷がつくる電界は，その導体板の両側で対称である．$x < 0$ および $x > H$ の領域における電界をそれぞれ求めよ．
(2) 導体板 A，B の間（$0 < x < H$）における電界を求めよ．
(3) 導体板 A を基準とした導体板 B の電位を求めよ．

図 1 の状態から，導体板 A，B を抵抗で接続した．十分な時間が経過した後は，図 2 に示すように導体板 A の面電荷密度は $\sigma'_{\mathrm{A}}$，導体板 B の面電荷密度は $\sigma'_{\mathrm{B}}$ となった．

(4) $\sigma'_{\mathrm{A}}$ および $\sigma'_{\mathrm{B}}$ を求めよ．
(5) 図 2 に示した断面積 $S$ の無限長の柱状領域に蓄えられた静電エネルギーを $W_{\mathrm{e}}$ とする．図 1 の状態から図 2 の状態に変化する過程での $W_{\mathrm{e}}$ の変化量 $\Delta W_{\mathrm{e}}$ を求めよ．

図 2 の状態から，図 3 に示すように，$x = h\ (0 < h < H/2)$ の位置に面電荷密度 $\sigma_{\mathrm{C}}$ で帯電した十分に広い導体板 C を挿入した．導体板 C の厚さおよび端効果は無視でき，導体板 A と B をつないでいる接続は導体板 C の電荷分布には影響を与えないものとする．

(6) 十分な時間が経過した後，導体板 C に働く単位面積あたりの力の大きさと向きを求めよ．

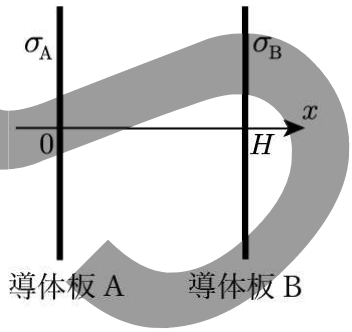



図 1

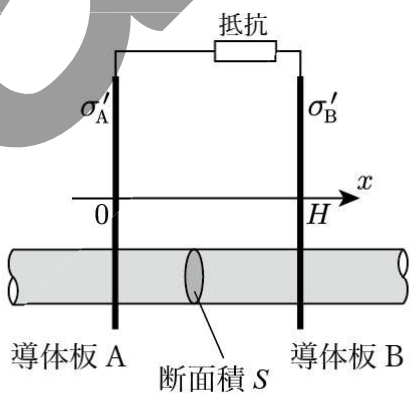



図 2

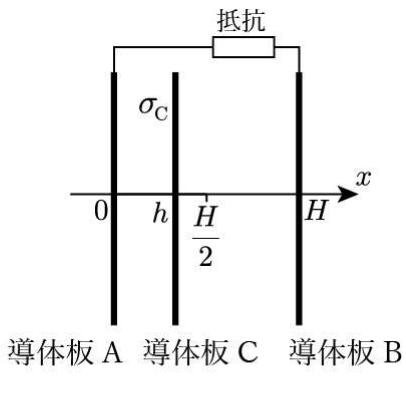



図 3

Sample

II. 以下の問に答えよ．

(1) 図 4 に示すように，真空中に置かれた半径 $a$（$> 0$），単位長さあたりの巻き数 $N$ の無限長ソレノイドに $+\theta$ 方向の定常電流 $I$ が流れている．ソレノイドの導線の太さは無視できるものとし，真空の透磁率を$\mu_0$とする．

(1-i) ソレノイドの中心軸からの距離を $r$ とする．ソレノイドの内側（$r < a$）および外側（$r > a$）における磁束密度を求めよ．

(1-ii) 中心軸から半径 $r$ の円内に含まれる磁束を求めよ．ただし，$z$ 軸正方向の磁束を正とする．

(1-iii) ソレノイドの単位長さあたりのインダクタンスを求めよ．

(1-iv) 一般に，曲面 $S$ を貫く磁束 $\Phi$ は，磁束密度 $\boldsymbol{B}$ と曲面の単位法線ベクトル $\boldsymbol{n}$ を用いて $\Phi = \int_S \boldsymbol{B}\cdot\boldsymbol{n}\,\mathrm{d}S$と表される．ソレノイド内外のベクトルポテンシャル $\boldsymbol{A} = (A_r, A_\theta, A_z)$ は $\theta$ 成分のみを持つことを利用し，$A_\theta$ を $r$ の関数として求めよ．

(2) 図 5 に示すように，内半径 $a$，外半径 $b$（$> a$）の無限長の円筒導体に$+\theta$ 方向に定常電流が流れている．円筒導体中の電流密度 $J$ は中心軸からの距離 $r$ に反比例し，$J(r) = J_0/r$ と表される（$J_0$ は正の定数）．

(2-i) 円筒導体の内側（$r < a$），内部（$a \le r \le b$），および外側（$r > b$）における磁界を求めよ．

(2-ii) 円筒導体の導電率を $\sigma$ とする．円筒導体の内側表面（$r = a$）および外側表面（$r = b$）におけるポインティングベクトル $\boldsymbol{S} = \boldsymbol{E} \times \boldsymbol{H}$ の大きさと向きを求めよ．なお，$\boldsymbol{E}$，$\boldsymbol{H}$ はそれぞれ電界と磁界を表す．

(2-iii) 円筒導体に，単位時間単位長さあたりに出入りするエネルギーを求めよ．

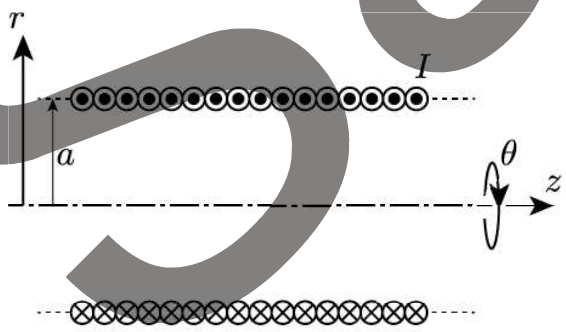



図 4

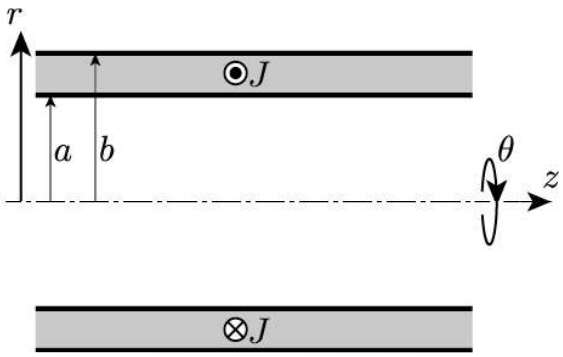



図 5

Sample

# 問題２

I. 図１に示す回路に関して，以下の問に答えよ．ここで$t$は時刻とする．図中の記号は凡例１に従う．解答において虚数単位は$j$とせよ．

(1) 問(1)では$R = 0$とする．

(1-i) スイッチ$S_1$を開き，スイッチ$S_2$を閉じているとき，角周波数$\omega$における A-B 間のアドミタンスを求めよ．

(1-ii) 十分長い間継続してスイッチ$S_1$と$S_2$をともに閉じているとする．電圧源から$\sqrt{2}E_1 \sin \omega t + \sqrt{2}E_2 \sin 3\omega t$で表される電圧を加える．キャパシタンス$C$が可変であるとき，回路に流れる電流$i$の実効値が最小となる場合の$C$の値を，$L$，$\omega$，$E_1$，$E_2$を用いて示せ．

(2) 十分長い間スイッチ$S_1$を閉じ，スイッチ$S_2$を開いているとする．$t < 0$では電圧源の電圧はゼロである．

(2-i) $t = 0$以降に電圧源から加えられる電圧が$E_3 \exp(-t/\tau)$で表されるとき，時刻$t(> 0)$において回路に流れる電流$i(t)$を求めよ．$\tau > 0$とする．

(2-ii) $t = 0$以降に電圧源から加えられる電圧が図２で表されるパルス波電圧であるとき，時刻$t(> 0)$において回路に流れる電流$i(t)$を求めよ．$T > 0$とする．

(3) 電圧源から直流電圧$E_5$を加える．十分長い間スイッチ$S_1$と$S_2$をともに閉じた後，$t = 0$でスイッチ$S_1$を開いたとする．

(3-i) $t > 0$でキャパシタ$C$の電荷$q$が満たす微分方程式を示せ．

(3-ii) 時刻$t(> 0)$における A-B 間の電圧$v(t)$を求めよ．

(3-iii) 問(3-ii)で求めた電圧の最大値は，$t = 0$にインダクタ$L$に蓄えられていたエネルギーを用いて求めることもできる．その理由を簡潔に述べよ．

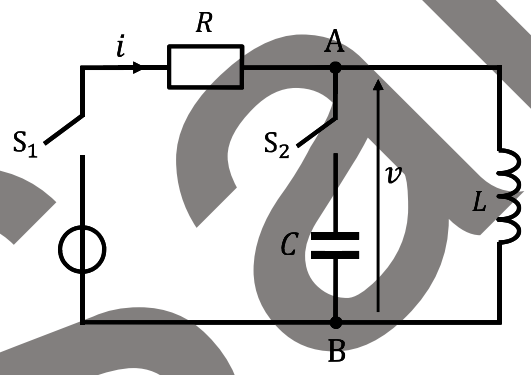



図１

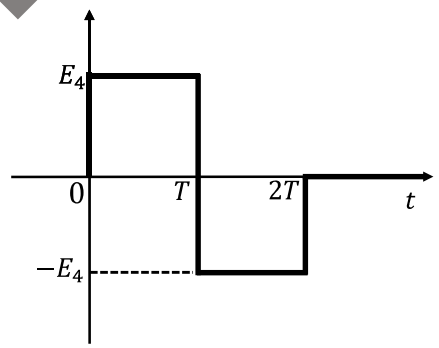
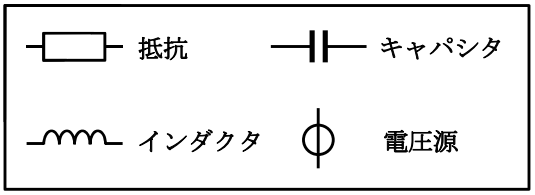



図２




凡例１

Sample

II. N 型 MOS トランジスタのドレイン電流$I_{\mathrm{D}}$は，ゲート・ソース間電圧$V_{\mathrm{GS}}$，ドレイン・ソース間電圧$V_{\mathrm{DS}}$，ボディ（基板）・ソース間電圧$V_{\mathrm{BS}}$の関数であるので，$V_{\mathrm{GS}}$，$V_{\mathrm{DS}}$，$V_{\mathrm{BS}}$がわずかに変化したときの$I_{\mathrm{D}}$の変化分$\Delta I_{\mathrm{D}}$は，以下の式で表される．

$$\Delta I_{\mathrm{D}} = g_{\mathrm{m}} v_{\mathrm{GS}} + g_{\mathrm{D}} v_{\mathrm{DS}} + g_{\mathrm{mb}} v_{\mathrm{BS}}$$

ただし，$v_{\mathrm{GS}}$，$v_{\mathrm{DS}}$，$v_{\mathrm{BS}}$は$V_{\mathrm{GS}}$，$V_{\mathrm{DS}}$，$V_{\mathrm{BS}}$の微小変化分とし，コンダクタンス$g_{\mathrm{m}}$，$g_{\mathrm{D}}$，$g_{\mathrm{mb}}$は以下を表すものとする．

$$g_{\mathrm{m}} = \frac{\partial I_{\mathrm{D}}}{\partial V_{\mathrm{GS}}}, \quad g_{\mathrm{D}} = \frac{\partial I_{\mathrm{D}}}{\partial V_{\mathrm{DS}}}, \quad g_{\mathrm{mb}} = \frac{\partial I_{\mathrm{D}}}{\partial V_{\mathrm{BS}}}$$

以下の問に答えよ．図中の記号は凡例２に従う．解答には凡例２の記号を使うこと．解答において虚数単位は$j$とせよ．

(1) ここでは MOS トランジスタの容量成分は考慮しないものとする．

(1-i) この N 型 MOS トランジスタの小信号等価回路を示せ．

(1-ii) 問(1-i)で示した小信号等価回路を用いて，図３に示す回路の小信号利得$v_{\mathrm{OUT}}/v_{\mathrm{IN}}$を求めよ．

以降の問では，N 型 MOS トランジスタの容量成分を考慮する．ゲート・ドレイン間の容量成分を$C_{\mathrm{GD}}$，ゲート・ソース間の容量成分を$C_{\mathrm{GS}}$，ドレイン・ボディ間の容量成分を$C_{\mathrm{DB}}$とする．

(2) MOS トランジスタが飽和領域で動作している場合，容量成分$C_{\mathrm{GD}}$は$C_{\mathrm{GS}}$と比較して小さい．この理由を簡単に説明せよ．

(3) 容量成分$C_{\mathrm{GD}}$を無視し，$C_{\mathrm{GS}}$と$C_{\mathrm{DB}}$のみを考慮するとき，図３の回路の小信号等価回路を示せ．

(4) 問(3)で示した小信号等価回路を用いて，小信号利得$v_{\mathrm{OUT}}/v_{\mathrm{IN}}$を正弦波電圧源の角周波数$\omega$の関数として求めよ．また，その周波数特性をボード線図で図示せよ．なお，ここでは入力電圧源の内部抵抗$R_{\mathrm{S}}$はゼロとする．

(5) 入力電圧源の内部抵抗$R_{\mathrm{S}}$がゼロでない場合，ボード線図が問(4)の結果からどのように変化するかを簡潔に説明せよ．

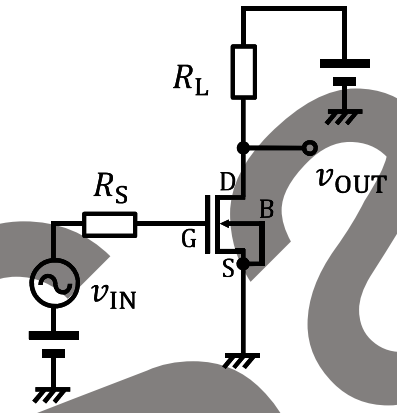
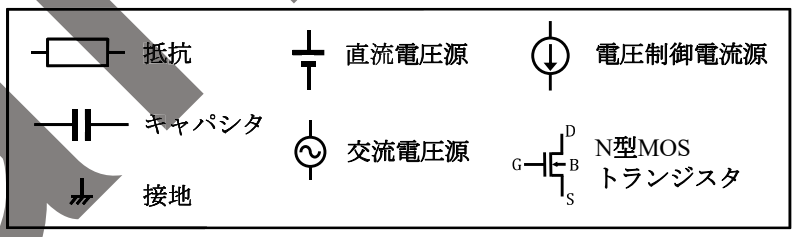



図３




凡例２

Sample

# 問題３

I. 情報理論に関する以下の問に答えよ．必要に応じて近似値として$\log_2 3 = 1.58$，$\log_2 5 = 2.32$，$\log_2 7 = 2.81$を用いてよい．

4 つの状態$\{K_0, K_1, K_2, K_3\}$を持つ無記憶情報源$K$について考える．各状態$K_0$，$K_1$，$K_2$，$K_3$をとる確率は$3/8$，$1/4$，$1/4$，$1/8$とする．

(1) 情報源$K$のエントロピー$H(K)$を求めよ．

(2) $K_0$，$K_1$，$K_2$，$K_3$をなるべく短い符号長になるように符号化する．その符号化の例を 1 つ示し，そのときの平均符号長を求めよ．また，この平均符号長と問(1)で求めたエントロピー$H(K)$との関係を簡潔に述べよ．

次に，4 つの状態$\{S_0, S_1, S_2, S_3\}$を持つマルコフ情報源を$S$とする．各試行で$S$は以下の式で与えられる遷移行列$T$に従って次の状態に遷移する．

$$T = \begin{bmatrix} p(S_0|S_0) & p(S_1|S_0) & p(S_2|S_0) & p(S_3|S_0) \\ p(S_0|S_1) & p(S_1|S_1) & p(S_2|S_1) & p(S_3|S_1) \\ p(S_0|S_2) & p(S_1|S_2) & p(S_2|S_2) & p(S_3|S_2) \\ p(S_0|S_3) & p(S_1|S_3) & p(S_2|S_3) & p(S_3|S_3) \end{bmatrix} = \begin{bmatrix} 1/3 & 2/3 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 \\ 1/3 & 2/3 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 \end{bmatrix}$$

(3) 情報源$S$の状態遷移図を示せ．

(4) 試行を十分な回数繰り返した後，それぞれの状態$S_0$，$S_1$，$S_2$，$S_3$になる確率$p_0$，$p_1$，$p_2$，$p_3$を求めよ．

(5) 情報源$S$のエントロピー$H(S)$を求めよ．

Sample

II. 信号処理に関する以下の問に答えよ．単位ステップ信号$u_s(t)$を次式で定義する．

$$u_s(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

時間$t$および角周波数$\omega$を実数，$j$を虚数単位とした時，連続時間信号$x(t)$のフーリエ変換$X(\omega)$を次式で定義する．

$$X(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

ここで，図１で示される信号$x_1(t)$を考える．$T > 0$とする．

(1) 信号$x_1(t)$を，単位ステップ信号$u_s(t)$を用いて表せ．
(2) ある線形時不変システムに，単位インパルス信号$\delta(t)$を入力した時の出力信号$f(t)$が次式で与えられる．

$$f(t) = \begin{cases} 0, & t < 0 \\ e^{-2t}, & t \geq 0 \end{cases}$$

この線形時不変システムに，信号$x_1(t)$を入力した時の出力信号を$g(t)$とする．この$g(t)$のフーリエ変換$\mathcal{F}[g(t)]$を求めよ．

次に，図２で示される信号$x_2(t)$を考える．$T > 0$とする．

(3) 信号$x_2(t)$を，単位ステップ信号$u_s(t)$を用いて表せ．
(4) 信号$x_2(t)$のフーリエ変換$\mathcal{F}[x_2(t)]$を求めよ．

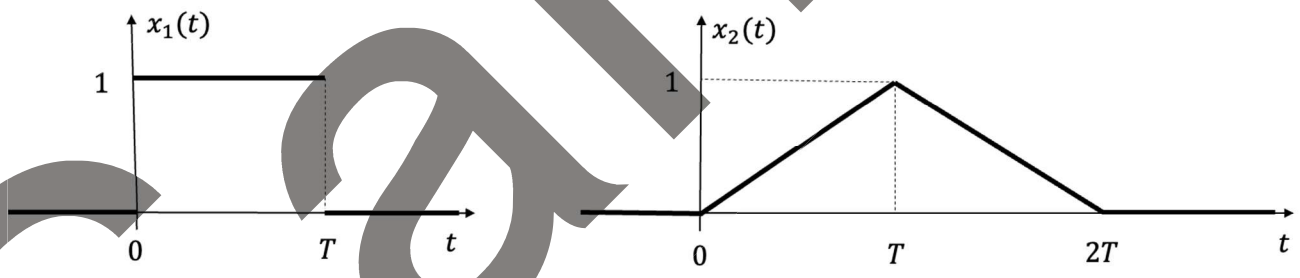


図１



図２

Sample

# 問題４

I. 論理回路に関する以下の問に答えよ．論理変数$x$，$y$に対する論理積(AND)，論理和(OR)，排他的論理和(XOR)，否定(NOT)をそれぞれ $x \cdot y$，$x + y$，$x \oplus y$，$\bar{x}$ と表す．D フリップフロップ(D-FF)の初期値は 0 とする．

(1) 2 つの D-FF からなる 4 進カウンタの真理値表を示せ．またその論理式を可能な限り簡単化した形で示せ．なお上位ビットと下位ビットのクロック$t$における値をそれぞれ$Q_1(t)$，$Q_0(t)$とし，次のクロックにおける値を$Q_1(t+1)$，$Q_0(t+1)$とすること．

(2) 図 1 に示す記号のみを用いて問(1)の 4 進カウンタの回路図を示せ．なお回路は可能な限り簡単化すること．

クロックごとに入力として 1 ビットの信号$x$を受け取り，一連の入力が特定のパターンに一致にしたときに 1 を，それ以外のときは 0 を 1 ビットの信号として出力する回路について考える．

(3) 図 2 に示す回路に，入力$x$として一連の 4 つの 1 ビットの信号を入力する．この回路は，一連の入力が順番に 0, 1, 0, 0 (0100)，同様に 0110，1010，1110 のいずれかであるときに 1 ビット信号$z$として 1 を，それ以外のときは 0 を出力する．図 2 中の[A]は$q_0$，$q_1$，$q_2$，$q_3$を入力として，$z$を出力する回路である．[A]に挿入する回路の論理式を可能な限り簡単な積和形で示せ．

(4) 2 つの D-FF による 4 つの状態を持つ順序回路で，一連の入力が 0110 であるときに 1 ビット信号$w$として 1 を出力する回路を考える．初期状態から 1 を出力するまでに遷移する状態$S$を順に$s0$，$s1$，$s2$，$s3$としてこの回路の状態遷移図を示せ．また表 1 はクロック$t$における状態を$S(t)$としたこの回路の状態遷移表である．表 1 を解答用紙にコピーしこの回路の状態遷移表を完成させよ．

(5) 問(4)の回路について，状態$S$に次のように符号を割り当てる：$s0 = 00$，$s1 = 01$，$s2 = 10$，$s3 = 11$．$S$の上位ビットを$Y_1$，下位ビットを$Y_0$とし，クロック$t$におけるそれぞれを$S(t)$，$Y_1(t)$，$Y_0(t)$とする．$Y_1(t+1)$，$Y_0(t+1)$，$w$のそれぞれを，$Y_1(t)$，$Y_0(t)$，$x$の論理式として可能な限り簡単化した形で示せ．

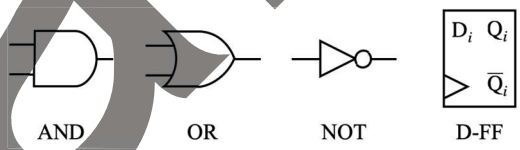



図 1

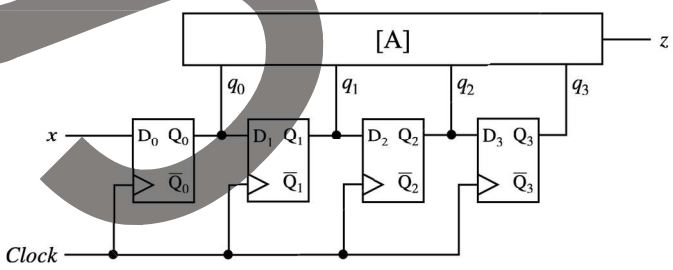



図 2

表 1

| Current State $S(t)$ | Next State $S(t+1)$ | | Output $w$ | |
|---|---|---|---|---|
| | $x = 0$ | $x = 1$ | $x = 0$ | $x = 1$ |
| $s0$ | | | | |
| $s1$ | | | | |
| $s2$ | | | | |
| $s3$ | | | | |

Sample

II. 木構造に'0'か'1'の文字からなる文字列（ビット列）を格納し，あるビット列がその木構造に存在するかどうかを判定する C 言語のプログラムを考える．木構造を構成するノードは Program 1 に示す構造体で表現される．長さ$m$のビット列$a = a_0 a_1 \ldots a_{m-1}$を木構造に挿入する操作は，深さ$i$のノードで$a_i$が'0'なら child[0]へ，$a_i$が'1'なら child[1]へ進み，深さ$m$のノードの変数 end を 1 にする．

(1) 図 3 は，空の木に 4 つのビット列"00"，"01"，"000"，"110"を挿入した状態を表している．二重丸のノードは変数 end が 1 であることを示している．図 3 の例に倣って空の木にビット列"1"，"01"，"001"，"111"，"0010"が挿入された状態を図示せよ．

(2) Program 2 は木に新しいビット列を格納する関数 insert()の再帰を用いた実装であり，Program 4 にあるように呼び出される．Program 2 中の new_node()は初期化された新しい node_t のオブジェクトを返す関数であり，char2bit()は引数の文字 c が'0'の場合は整数 0 を，c が'1'の場合は整数 1 を返す関数である． [A], [B]に入るコードを記述せよ．なお入力となるビット列の最後には Null 文字'\0'が付与されている．

(3) Program 3 は与えられたビット列が木の中に存在するときは整数 1 を，そうでないときは 0 を返す関数 exists()の再帰を用いない実装であり，プログラム 4 にあるように呼び出される．[C]，[D]，[E]に入るコードを記述せよ．

(4) $n$個のビット列が格納された木があるとき，ある長さ$m$のビット列が木の中に存在するかを判定する操作の時間計算量を示せ．また同じ操作を単純な二分探索木で実現する場合と比較して，この木構造が構造上有利な点を簡潔に説明せよ．

(5) この木構造を用いて，任意のビット列の集合を辞書順にソートする．このときの時間計算量を示せ．

```
/* Program 1 */
typedef struct node {
    struct node *child[2];
    int end;
} node_t;
```

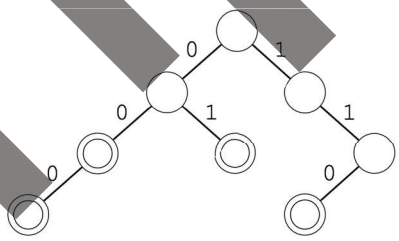



図 3

```
/* Program 2 */
node_t *new_node(void);
int char2bit(char c);

void _insert(node_t *node, char str[], int i) {
    if (str[i] == '\0') {
        [A]
        return;
    }
    int j = char2bit(str[i]);
    if (node->child[j] == NULL) {
        node->child[j] = new_node();
    }
    [B]
}

void insert(node_t *node, char str[]) {
    _insert(node, str, 0);
}
```

```
/* Program 3 */
int exists(node_t *node, char str[]) {
    int i;
    for (i = 0; node != NULL; i++) {
        if (str[i] == '\0') {
            [C]
        } else if (str[i] == '0') {
            [D]
        } else if (str[i] == '1') {
            [E]
        }
    }
    return 0;
}
```

```
/* Program 4 */
int main(void) {
    node_t root = {};
    insert(&root, "01");
    insert(&root, "110");
    return exists(&root, "010");
}
```

Sample

# 問題５

I.　一次元系の電子の波動関数と量子状態について以下の問に答えよ．電子は$x$方向にのみ運動できるものとする．$m$は電子の質量，$\hbar$はプランク定数$h$を$2\pi$で割った定数，$k_{\mathrm{B}}$はボルツマン定数，$i$は虚数単位である．

(1) 固有エネルギー$E$をもつ単一電子の波動関数$\psi(x)$は，時間に依存しないシュレディンガー方程式

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}+V(x)\psi(x)=E\psi(x)$$

に従う．図１に示した

$$V(x)=\begin{cases}+\infty & (x\le 0)\\ V_0 & (0<x<L)\\ +\infty & (x\ge L)\end{cases}$$

のポテンシャル$V(x)$の中に存在する電子の固有状態について考える．

(1-i)　固有エネルギー$E$をもつ電子の波動関数は$0<x<L$において，$A$，$B$，$\phi$を定数として$\psi(x)=A\sin(Bx+\phi)$の形で表せることを示し，$B$を$m$，$\hbar$，$V_0$，$E$を用いて示せ．

(1-ii)　境界条件を用いて取りうる固有エネルギー$E$と定数$\phi$を求めよ．

(1-iii)　次の文章の［ I ］と［ II ］に入る適切な表現を記せ．

「規格化された波動関数の絶対値の二乗は［ I ］関数と呼ばれる．［ I ］関数を$x_1\le x\le x_2$の区間で積分した値は［ II ］という物理的な意味をもつ．」

$V(x)$の中に複数の電子が存在する系を考える．固有エネルギー$E$は単一電子系から変わらないとする．電子は熱平衡状態にあり，絶対温度$T$におけるフェルミ準位$E_{\mathrm{F}}$のフェルミ・ディラック分布に従う．

(1-iv)　固有エネルギー$E$の状態が電子に占有される確率と，固有エネルギー$E$をもつ電子数の期待値を求めよ．

(2) 自由空間中の電子の運動量について考える．運動量演算子は$\hat{p}=-i\hbar(d/dx)$で与えられる．

(2-i)　次式で与えられる波動関数$\varphi(x)$の電子を考える．この電子の運動量の期待値$\langle\hat{p}\rangle$を求めよ．正の整数$N$，実部が正の複素数$\beta$に対して$\int_0^\infty x^N e^{-\beta x}dx=N!/\beta^{N+1}$の関係を用いて良い．

$$\varphi(x)=\begin{cases}0 & (x<0)\\ 2a^{1.5}xe^{-(a+ib)x} & (x\ge 0)\end{cases}\quad ;a\text{と}b\text{は正の実数}$$

(2-ii)　電子の運動量を観測したところ，ある値$p_1$が得られた．この運動量の観測によって波動関数がどのように変化するかを説明し，観測直後の波動関数を求めよ．

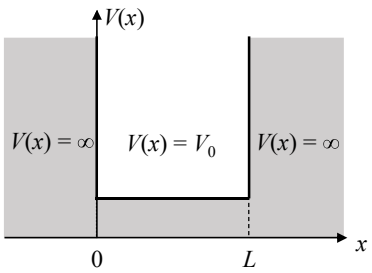



図１

Sample

II. 金属・酸化物・半導体（MOS）構造を考える．フラットバンド状態におけるエネルギーバンド図を図２に示す．半導体は十分に厚く，不純物濃度$N_A$の非縮退P型シリコンである．半導体中の不純物は完全にイオン化している．また，禁制帯中や酸化物との界面に欠陥準位は存在しない．シリコンの真性キャリア密度を$n_i(\ll N_A)$，比誘電率を$\varepsilon_s$とする．酸化物は厚さ$t_{ox}$と比誘電率$\varepsilon_{ox}$ $(<\varepsilon_s)$をもつ．シリコン側に対して金属側に電圧$V_g$が印加されている．絶対温度を$T$，真空における誘電率を$\varepsilon_0$，ボルツマン定数を$k_B$，電荷素量を$q$とする．以下の問に答えよ．

(1) フラットバンド状態において，P型シリコンにおける正孔密度$p_0$と電子密度$n_0$を表せ．

(2) フラットバンド状態において，P型シリコンにおけるフェルミ準位$E_F$を，真性フェルミ準位$E_i$を用いて表せ．

(3) 印加する電圧$V_g$によってシリコンは蓄積状態，空乏状態，反転状態になる．これら3つの状態のMOS構造のバンド図を模式的に図示せよ．またそれぞれの状態となる電圧$V_g$の範囲を示せ．シリコンのフェルミ準位$E_F$と真性フェルミ準位$E_i$の位置も図中に示すこと．ここで，MOS構造のフラットバンド電圧を$V_{FB}$，しきい値電圧を$V_{TH}$とする．

(4) 空乏近似を用いて空乏状態を考える．シリコンの表面ポテンシャルが$\phi_s$であるとき，空乏層幅$W_D$，金属上の面電荷密度$\sigma_M$および酸化物にかかる電圧$V_{ox}$を，$\phi_s$の関数として求めよ．

(5) 空乏層幅$W_D$の空乏状態における，MOS構造の単位面積当たりの容量$C = d\sigma_M/dV_g$を，$\varepsilon_0$, $\varepsilon_{ox}$, $\varepsilon_s$, $t_{ox}$, $W_D$を用いて表せ．

(6) 絶対温度$T$が低くなると次の物理量が高くなるか低くなるかを記せ．それぞれの理由を1-2行で述べよ．ここで，各層の誘電率と厚さ，不純物のイオン化率，金属の仕事関数は変わらないものとする．

(6-i) フラットバンド状態におけるP型シリコンの電子密度$n_0$

(6-ii) フラットバンド状態におけるP型シリコンのフェルミ準位$E_F$

(6-iii) フラットバンド電圧$V_{FB}$

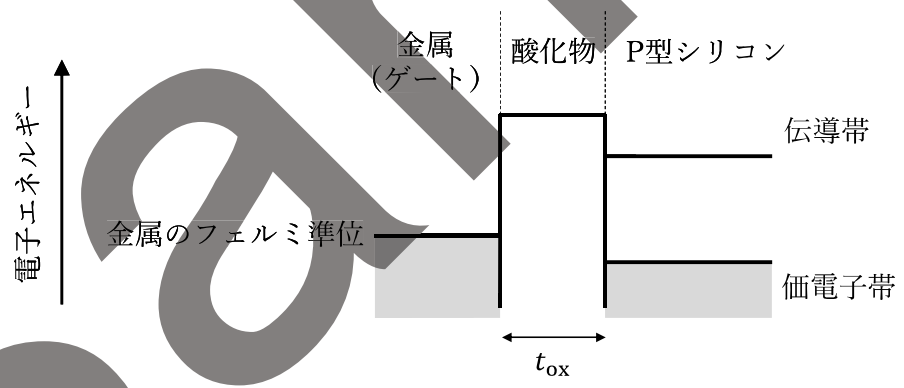



図２

Sample

# 問題 6

I. 図 1 で示す伝達特性を持つ制御対象がある．$\tau$, $J$は正の定数, $s$はラプラス変換の変数である．この制御対象に対し，図 2 のように，$K_P$をゲインとする比例制御器を用いたフィードバック制御を行う．$U(s)$，$Y(s)$，$D(s)$，$Y^*(s)$は，各々，入力信号$u(t)$，出力信号$y(t)$，外乱$d(t)$，指令値$y^*(t)$の時間波形のラプラス変換である．以下の問に答えよ．

(1) 図 1 において，入力から出力への伝達関数$G_0(s) = \frac{Y(s)}{U(s)}$を求めよ．

(2) 図 2 において，出力$Y(s)$の指令値$Y^*(s)$から$Y(s)$への閉ループ伝達関数$G_1(s) = \frac{Y(s)}{Y^*(s)}$を求めよ．

(3) 図 2 において，外乱$D(s)$から出力$Y(s)$への閉ループ伝達関数$G_2(s) = \frac{Y(s)}{D(s)}$を求めよ．

これ以降の問題では，$\tau = J = 1.0$とする．

(4) 問(1)の$G_0(s)$のボード線図の概形を描け．また，$K_P$の値に応じて図 2 の閉ループシステムの安定性がどう変化するか，このボード線図を用いて説明せよ．

(5) 図 2 において$K_P = \frac{1}{8}$，$d(t) = 0$とする．指令値$y^*(t)$が単位ステップ関数であるとき，出力$y(t)$を求めよ．さらに，$y(t)$の概形を描け．

(6) 図 2 において$K_P = \frac{1}{8}$，$y^*(t) = 0$とする．外乱$d(t)$が単位ステップ関数であるとき，$y(t)$の概形を描き，$y(t)$の最終値を求めよ．

(7) $K_P$を大きくしていくと， 問(5), (6)の単位ステップ応答の波形や最終値が各々どのように変化するか，特に応答の振動周波数，減衰や安定性にどのような影響があるかを，必要に応じ式や図を用いて説明せよ．

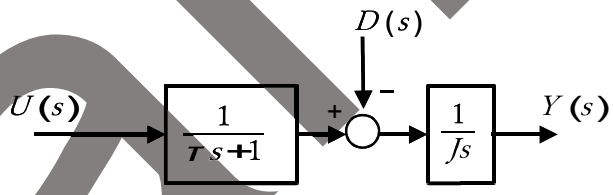



図 1

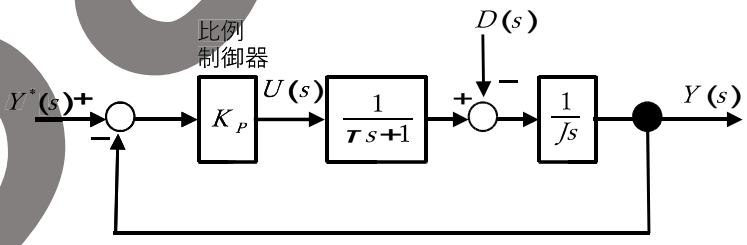



図 2

Sample

II. 永久磁石を界磁とする他励直流機がある．界磁磁束は一定であり，電機子抵抗は$R_{\mathrm{a}} = 0.5\ \Omega$ の一定値である．簡単のため，ブラシ・整流子間の電圧降下や電機子反作用は無視できるものとする．また，鉄損，機械損など電機子損失以外の損失要素も十分に小さく無視できるものとする．円周率を$\pi$とする．以下の問に答えよ．

(1) 図 3 のように電機子端子に$V_{\mathrm{a}} = 100\ \mathrm{V}$ の電源を接続し，この直流機を電動機として運転したところ，電機子電流$I_{\mathrm{a}} = 20\ \mathrm{A}$ であり，電機子の回転数は 1 分あたり 1500 回転であった．この時の，内部起電力$E_0$ [V]と回転子の回転角速度$\omega_{\mathrm{mot}}$ [rad/s]を求めよ．それらを基に電動機出力 $P_{\mathrm{out}}$ [W]と電動機トルク$T_{\mathrm{mot}}$ [Nm]を求めよ．そして，電動機効率$\eta_{\mathrm{mot}}$を求めよ．

(2) 図 4 のように，問(1)と同じ直流機の電機子端子に負荷抵抗$R_{\mathrm{L}} = 5.0\ \Omega$ を接続して，電機子に回転力を加え回転させた．その結果，電機子電圧$V_{\mathrm{a}}' = 100\ \mathrm{V}$ を得た．このときの回転角速度$\omega_{\mathrm{gen}}$ [rad/s]と発電機効率$\eta_{\mathrm{gen}}$を求めよ．

(3) 問(1)，(2)で扱ったエネルギー変換やその効率に関する考察を踏まえ，エネルギー利用における電気自動車の利点を簡潔に説明せよ．

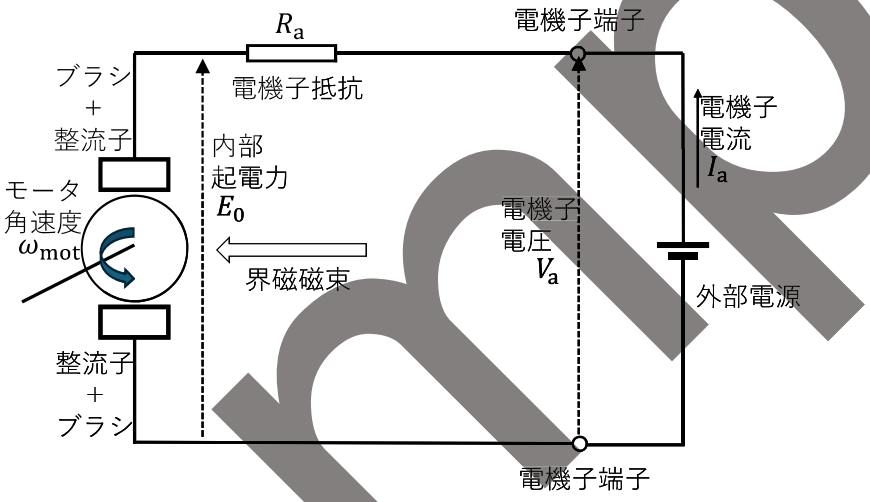



図 3

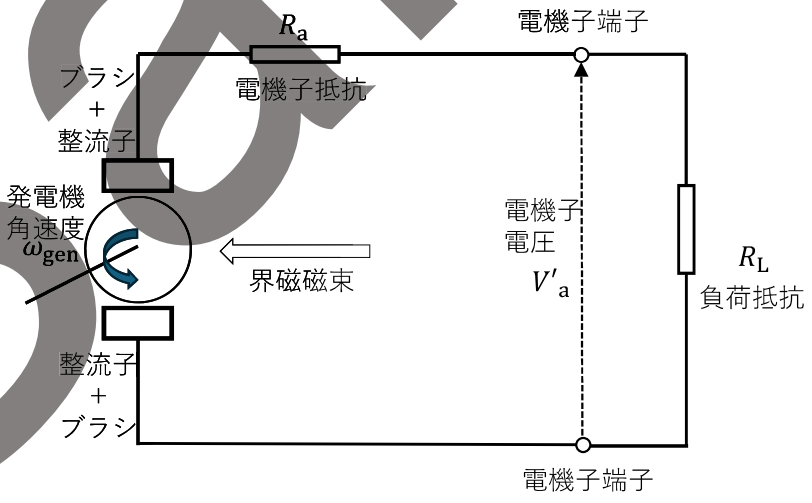



図 4

Sample

## Problem 1

I. Two sufficiently-large conducting plates are placed in vacuum. As shown in Fig. 1, conducting plates A and B are located parallel to each other at positions $x = 0$ and $x = H\ (> 0)$, and they are charged with uniform surface charge densities of $\sigma_{\mathrm{A}}$ and $\sigma_{\mathrm{B}}$, respectively. The thickness and the edge effect of the conducting plates can be neglected, and the electric field and the charge density in the conducting plates have only the $x$-component. Let $\varepsilon_0$ be the vacuum permittivity. Answer the following questions.

(1) The electric field generated by the charges on each conducting plate is symmetric on both sides of the corresponding conducting plate. Find the electric field in the regions of $x < 0$ and $x > H$.

(2) Find the electric field between conducting plates A and B ($0 < x < H$).

(3) Find the electric potential of conducting plate relative to conducting plate A.

From the condition in Fig. 1, conducting plates A and B are connected with a resistor. After sufficient time, the surface charge densities of conducting plates A and B reach $\sigma'_{\mathrm{A}}$ and $\sigma'_{\mathrm{B}}$, respectively, as shown in Fig. 2.

(4) Find $\sigma'_{\mathrm{A}}$ and $\sigma'_{\mathrm{B}}$.

(5) Let $W_{\mathrm{e}}$ be the electrostatic energy stored in the infinitely long columnar region with cross-sectional area $S$, as indicated in Fig. 2. Find $\Delta W_{\mathrm{e}}$, which represents the change in $W_{\mathrm{e}}$ from the condition in Fig. 1 to that in Fig. 2.

From the condition in Fig. 2, sufficiently-large conducting plate C with surface charge density $\sigma_{\mathrm{C}}$ is inserted at a position $x = h\ (0 < h < H/2)$, as shown in Fig. 3. The thickness and the edge effect of conducting plate C can be neglected, and the insertion between conducting plates A and B does not affect the surface charge distribution on conducting plate C.

(6) Find the magnitude and direction of the force per unit area acting on conducting plate C after sufficient time has passed.

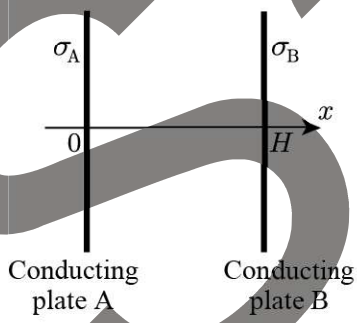



Fig. 1

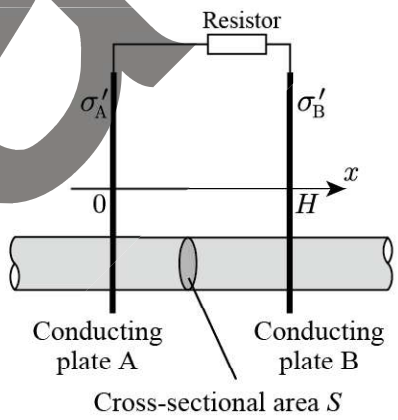



Fig. 2

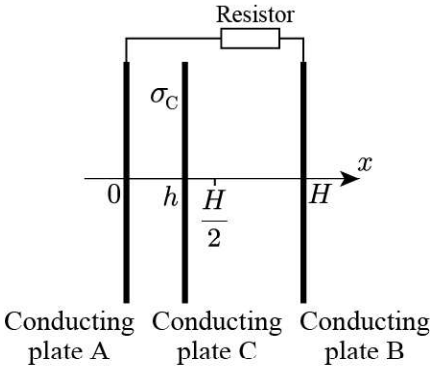



Fig. 3

Sample

II. Answer the following questions.

(1) As shown in Fig. 4, a constant current $I$ flows in the $+\theta$-direction in an infinitely-long solenoid with the radius $a\ (>0)$ and the number of turns per unit length $N$ placed in vacuum. The thickness of the solenoid wire can be neglected. Let $\mu_0$ be the vacuum permeability.

(1-i) Let $r$ be the distance from the central axis of the solenoid. Find the magnetic flux density in the radially inner region $(r < a)$ and the outer region $(r > a)$ of the solenoid.

(1-ii) Find the magnetic flux contained within a circle of radius $r$ from the central axis. Note that the magnetic flux in the positive direction of the $z$-axis is defined as positive.

(1-iii) Find the inductance per unit length of the solenoid.

(1-iv) Generally, the magnetic flux $\Phi$ passing through a surface $S$ is expressed as $\Phi = \int_S \boldsymbol{B}\cdot\boldsymbol{n}\,\mathrm{d}S$, where $\boldsymbol{B}$ is the magnetic flux density and $\boldsymbol{n}$ is the unit normal vector of the surface. Considering that the vector potential $\boldsymbol{A} = (A_r, A_\theta, A_z)$ has only the $\theta$-component both inside and outside the solenoid, find $A_\theta$ as a function of $r$.

(2) As shown in Fig. 5, a constant current flows in the $+\theta$-direction in an infinitely-long thick conducting cylinder with the inner radius $a$ and the outer radius $b\ (>a)$. The current density $J$ in the thick conducting cylinder is inversely proportional to the distance $r$ from the central axis and is expressed as $J(r) = J_0/r$, where $J_0$ is a positive constant.

(2-i) Find the magnetic field in the radially inner region $(r < a)$, the inside region $(a \le r \le b)$, and the radially outer region $(r > b)$ of the thick conducting cylinder.

(2-ii) Let $\sigma$ be the conductivity of the thick conducting cylinder. Find the magnitude and direction of the Poynting vector $\boldsymbol{S} = \boldsymbol{E}\times\boldsymbol{H}$ at the inner surface $(r = a)$ and the outer surface $(r = b)$ of the thick conducting cylinder. Here, $\boldsymbol{E}$ and $\boldsymbol{H}$ represent the electric field and the magnetic field, respectively.

(2-iii) Find the energy transfer per unit length, per unit time to/from the thick conducting cylinder.

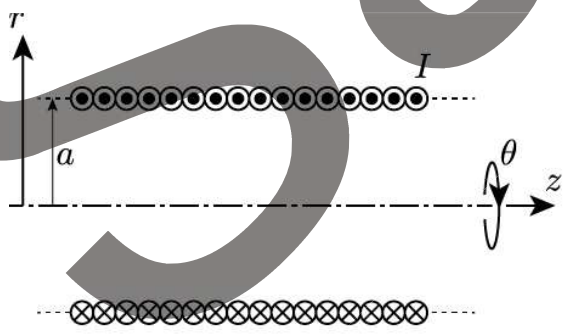



Fig. 4

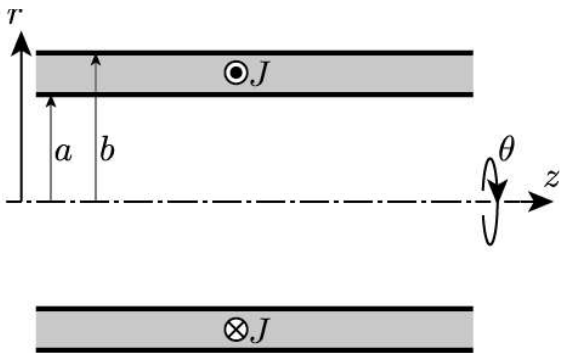



Fig. 5

Sample

## Problem 2

I. Answer the following questions on the electrical circuit shown in Fig. 1. Here, $t$ is time. Symbols in the figures are defined in Legend 1. Use $j$ as the imaginary unit in the answer.

(1) In Question (1), let $R = 0$.

(1-i) The switch $S_1$ is opened and the switch $S_2$ is closed. Find the admittance between the nodes A and B at an angular frequency $\omega$.

(1-ii) Both the switches $S_1$ and $S_2$ have been closed for a sufficiently long time. The voltage $\sqrt{2}E_1 \sin \omega t + \sqrt{2}E_2 \sin 3\omega t$ is applied from the voltage source S p i th t th c pacitance $C$ is variable, express $C$ using $L$, $\omega$, $E_1$, and $E_2$ when the effective value of the current $i$ flowing through the circuit becomes minimum.

(2) The switch $S_1$ has been closed and the switch $S_2$ has been opened for a sufficiently long time. When $t < 0$, the voltage applied from the voltage source is zero.

(2-i) When the voltage applied from the voltage source after $t = 0$ is expressed as $E_3 \exp(-t/\tau)$, find the current $i(t)$ flowing through the circuit at the time $t(> 0)$. Let $\tau > 0$.

(2-ii) When the voltage applied fro e .age source after $t = 0$ is the pulse wave voltage shown in Fig. 2, find the current $i(t)$ flowing through the circuit at the time $t(> 0)$. Let $T > 0$.

(3) The DC voltage $E_5$ is applied from the voltage source. After both the switches $S_1$ and $S_2$ have been closed for a sufficiently long time th switch $S_1$ is opened at $t = 0$.

3-i) th d ffere ti l equation which the electric charge $q$ of the capacitor $C$ satisfies for $t > 0$.

(3-ii) Find the voltage $v(t)$ between the nodes A and B at the time $t(> 0)$.

(3-iii) he m ximum value of the voltage derived in Question (3-ii) can also be obtained from the energy stored in t e i ductor $L$ at $t = 0$. Explain the reason briefly.

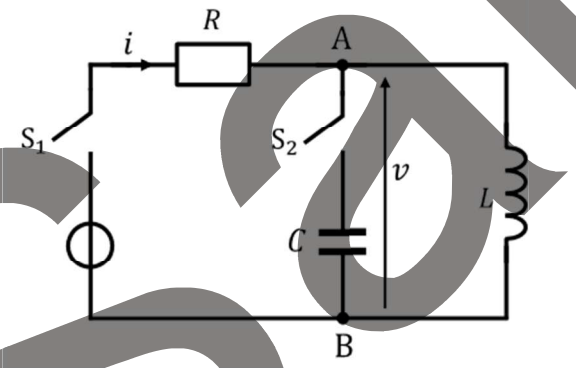



Fig. 1

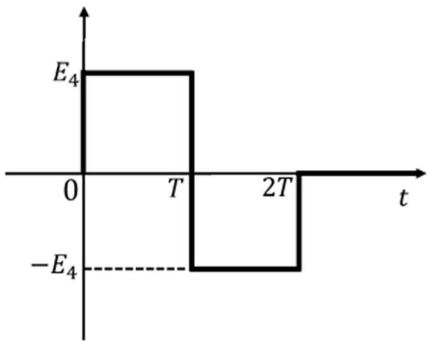
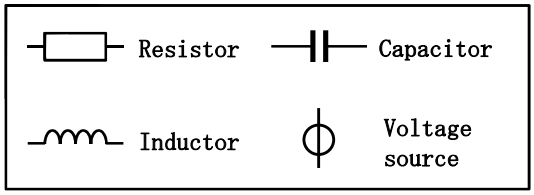



Fig. 2




Legend 1

Sample

II. The drain current $I_D$ of the N-type MOS transistor is the function of the voltage $V_{GS}$ between the gate and the source, the voltage $V_{DS}$ between the drain and the source, and the voltage $V_{BS}$ between the body (substrate) and the source. Hence, the current difference $\Delta I_D$ when $V_{GS}$, $V_{DS}$ and $V_{BS}$ change slightly is expressed as

$$\Delta I_D = g_m v_{GS} + g_D v_{DS} + g_{mb} v_{BS},$$

where $v_{GS}$, $v_{DS}$, and $v_{BS}$ are the small differences of $V_{GS}$, $V_{DS}$ and $V_{BS}$, and the conductance $g_m$, $g_D$, and $g_{mb}$ denote as the followings, respectively,

$$g_m = \frac{\partial I_D}{\partial V_{GS}}, \quad g_D = \frac{\partial I_D}{\partial V_{DS}}, \text{ and } \quad g_{mb} = \frac{\partial I_D}{\partial V_{BS}}.$$

Answer the following questions. Symbols in the figures are defined in Legend 2. Use the symbols of Legend 2 in the answer. Use $j$ as the imaginary unit in the answer.

(1) The capacitive components of the MOS transistor are ignored in this question.
  (1-i) Draw a small-signal equivalent circuit of the N-type MOS transistor.
  (1-ii) Using the small-signal equivalent circuit of Question (1-i), find the small signal gain of the circuit shown in Fig. 3, $v_{OUT}/v_{IN}$.

In the following questions, we consider the capacitive components of the N-type MOS transistor: the capacitive component $C_{GD}$ between the gate and the drain, the capacitive component $C_{GS}$ between the gate and the source, and the capacitive component $C_{DB}$ between the drain and the body.

(2) Explain briefly the reason why the capacitive component $C_{GD}$ is smaller than $C_{GS}$ when the MOS transistor is operated in the saturation region.

(3) When we ignore the capacitive component $C_{GD}$ and consider only $C_{GS}$ and $C_{DB}$, draw a small-signal equivalent circuit for the circuit in Fig. 3.

(4) Using the small-signal equivalent circuit of Question (3), find the small signal gain $v_{OUT}/v_{IN}$ as a function of the angular frequency $\omega$ of the sinusoidal voltage source. Then, draw the Bode diagram of the frequency characteristics of $v_{OUT}/v_{IN}$. Here, assume that the internal resistance $R_S$ of the input voltage source is zero.

(5) When the internal resistance $R_S$ of the input voltage source is non-zero, explain briefly how the Bode diagram changes from the result of Question (4).

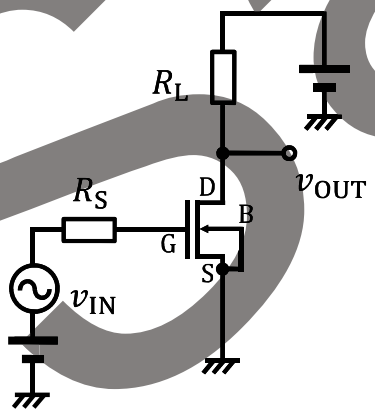
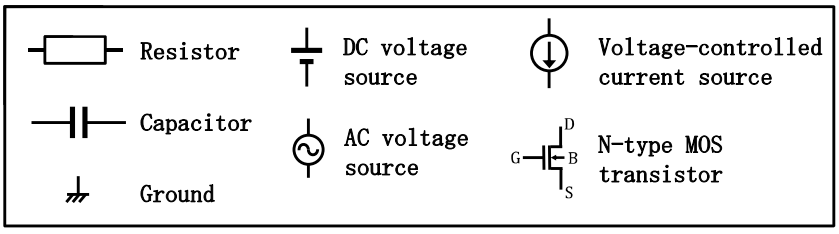



Fig. 3




Legend 2

Sample

## Problem 3

I. Answer the following questions on information theory. You may use the following approximations, when necessary: $\log_2 3 = 1.58$, $\log_2 5 = 2.32$, and $\log_2 7 = 2.81$.

Let $K$ be a memoryless information source that has four states $\{K_0, K_1, K_2, K_3\}$. The probability of each state $K_0$, $K_1$, $K_2$, and $K_3$ equals to $3/8$, $1/4$, $1/4$, and $1/8$, respectively.

(1) Find the entropy of the information source $K$, $H(K)$.

(2) We encode $K_0$, $K_1$, $K_2$, and $K_3$ with code words as short as possible. Show an example of code words and find its average length. Also briefly describe relationship between the average length of the code words and the entropy $H(K)$ obtained in Question (1).

Next, let $S$ be a Markov information source that has four states $\{S_0, S_1, S_2, S_3\}$. For each trial, $S$ makes a transition to a next state according to the transition probability matrix $T$ given by the following equation:

$$T = \begin{bmatrix} p(S_0|S_0) & p(S_1|S_0) & p(S_2|S_0) & p(S_3|S_0) \\ p(S_0|S_1) & p(S_1|S_1) & p(S_2|S_1) & p(S_3|S_1) \\ p(S_0|S_2) & p(S_1|S_2) & p(S_2|S_2) & p(S_3|S_2) \\ p(S_0|S_3) & p(S_1|S_3) & p(S_2|S_3) & p(S_3|S_3) \end{bmatrix} = \begin{bmatrix} 1/3 & 2/3 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 \\ 1/3 & 2/3 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 \end{bmatrix}.$$

(3) Draw the state transition diagram of the information source $S$.

(4) Find the probabilities $p_0$, $p_1$, $p_2$, and $p_3$ of being in the states $S_0$, $S_1$, $S_2$, and $S_3$, respectively, after a sufficient number of trials.

(5) Find the entropy of the information source $S$, $H(S)$.

Sample

II. Answer the following questions on signal processing. Here, the unit step signal $u_s(t)$ is given by

$$u_s(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}.$$

The Fourier transform $X(\omega)$ of a continuous-time signal $x(t)$ is given by

$$X(\omega) = \mathcal{F}[x(t)] = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt\,,$$

where time $t$ and angular frequency $\omega$ are real numbers, and $j$ is the imaginary unit.

Now, we consider a signal $x_1(t)$ shown in Fig. 1. Let $T > 0$.

(1) Express the signal $x_1(t)$ by using the unit step signal $u_s(t)$.

(2) When the unit impulse signal $\delta(t)$ is input to a linear time-invariant system, the output signal $f(t)$ is given by

$$f(t) = \begin{cases} 0, & t < 0 \\ e^{-2t}, & t \geq 0 \end{cases}.$$

Let $g(t)$ be the output signal of the linear time-invariant system with the input signal $x_1(t)$. Find the Fourier transform of the output signal $g(t)$, $\mathcal{F}[g(t)]$.

Next, we consider a signal $x_2(t)$ shown in Fig. 2. Let $T > 0$.

(3) Express the signal $x_2(t)$ by using the unit step signal $u_s(t)$.

(4) Find the Fourier transform of the signal $x_2(t)$, $\mathcal{F}[x_2(t)]$.

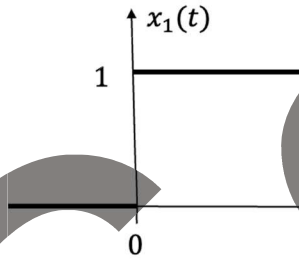



Fig. 1

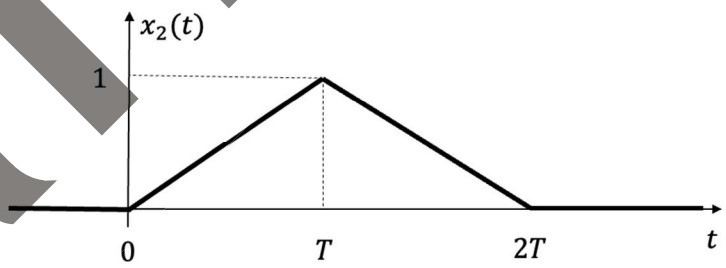



Fig. 2

Sample

## Problem 4

I. Answer the following questions on logic circuits. Logical conjunction (AND), logical disjunction (OR), exclusive disjunction (XOR), and negation (NOT) for logical variables $x$ and $y$ should be represented with $x \cdot y$, $x + y$, $x \oplus y$, and $\bar{x}$, respectively. The initial value of D flip-flop (D-FF) is 0.

(1) Write down the truth table of a quaternary counter circuit h s. so i a g cal formula of the quaternary counter. The logical formula must be simplified as much as possible. Note that the values of higher and lower bits at clock $t$ should be represented with $Q_1(t)$ and $Q_0(t)$, and the values at the next clock should be $Q_1(t+1)$ and $Q_0(t+1)$, respectively.

(2) ra a c rc i f th qu tern ry ount r i Question (1 y u i only e sy .ıbols in Fig. 1. The circuit must be simplified as much as possible.

Let us consider circuits that receive 1-bit signal $x$ on every clock. The circuits output 1 when a series of the 1-bit input matches specific patterns, and output 0, otherwise.

(3) c r it n ig ec i e q e e o 1- t .ıputs to $x$. The circuit outputs 1 to 1-bit signal $z$ when the series of inputs matches one of the sequences of 0, 1, 0, 0 (0100), 0110, 1010, or 1110. Otherwise, it outputs 0. [A] in Fig. 2 is a circuit with inputs $q_0$, $q_1$, $q_2$, and $q_3$, and the output $z$. Find the logical formula of the circuit that should be in [A] in Fig. 2 in e most simplified sum-of-products form.

(4) Consider a se uentia i u t having four states with two D-FFs. The circuit outputs 1 as 1-bit signal $w$ when the series of 1-bit input is 0110. Draw the state transition diagram of this circuit. Note that the state $S$ has the four states from the initial state until the output of 1. Define the four states in order as $s0$, $s1$, $s2$, and $s3$. Also, Table 1 is the state transition table of this circuit where the state $S$ at clock $t$ is represented with $S(t)$. Copy Table 1 to your answer sheet and complete the state transition table of this circuit.

(5) Assign codes to the state $S$ of the circuit in Question (4) as follows: $s0 = 00$, $s1 = 01$, $s2 = 10$, $s3 = 11$. The higher and lower bits of $S$ should be represented with $Y_1$ and $Y_0$, and those at clock $t$ should be $S(t)$, $Y_1(t)$, and $Y_0(t)$, respectively. Find $Y_1(t+1)$, $Y_0(t+1)$, and $w$ as logical formulae of $Y_1(t)$, $Y_0(t)$, and $x$. The logical formulae must be simplified as much as possible.

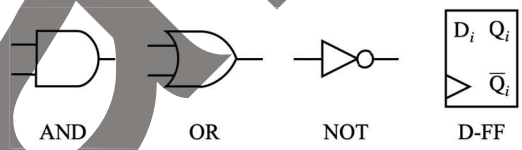



Fig. 1

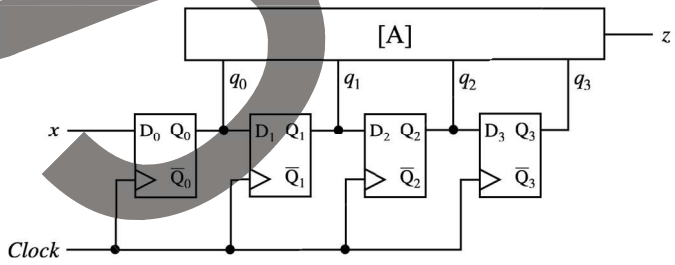



Fig. 2

Table 1

| Current State $S(t)$ | Next State $S(t+1)$ | | Output $w$ | |
|---|---|---|---|---|
| | $x = 0$ | $x = 1$ | $x = 0$ | $x = 1$ |
| $s0$ | | | | |
| $s1$ | | | | |
| $s2$ | | | | |
| $s3$ | | | | |

Sample

II. Consider a C language program that stores bit-strings, a series of characters of `'0'` and `'1'`, into a tree structure and searches the tree to determine whether a query bit-string exists in the tree or not. Each node of the tree is represented by a structure shown in Program 1. Inserting a bit-string $a = a_0 a_1 \ldots a_{m-1}$ of length $m$ to the tree is as follows: the program moves into `child[0]` when $a_i$ is `'0'`, or moves into `child[1]` when $a_i$ is `'1'` on a node of depth $i$, and a variable `end` of a node of depth $m$ is set to 1.

(1) Figure 3 exemplifies the tree where four bit-strings, `"00"`, `"01"`, `"000"`, and `"110"` are inserted into an empty tree. Double circle nodes indicate that their variables `end` are 1. Draw the resultant tree inserting bit-strings `"1"`, `"01"`, `"001"`, `"111"`, and `"0010"` to an empty tree by following the example shown in Fig. 3.

(2) Program 2 is a recursive implementation of a function `insert()` that inserts a bit-string into a tree. It is called as shown in Program 4. `new_node()` defined in Program 2 is a function that returns an initialized new object of `node_t`, and `char2bit()` is a function that returns an integer 0 when the argument character `c` is `'0'`, or an integer 1 when `c` is `'1'`. Write codes that should be in the blanks, [A] and [B]. Note that input bit-strings are terminated by the null character `'\0'`.

(3) Program 3 is a non-recursive implementation of a function `exists()` that returns 1 when a query bit-string exists in the tree, and 0, otherwise. It is called as shown in Program 4. Write codes that should be in the blanks, [C], [D], and [E].

(4) Suppose a tree where $n$ bit-strings are inserted. Find the time complexity for an operation of determining whether a given bit-string of length $m$ exists or not in the tree. Also, briefly explain the structural advantag ( ) f h tre str t re o p d witn one that implements the same operation by a basic binary search tre .

(5) Find the time complexity for sorting a set of arbitrary bit-strings in lexicographical order by using this tree structure.

```
/* Program 1 */
typedef struct node {
    struct node *child[2];
    int end;
} node_t;
```

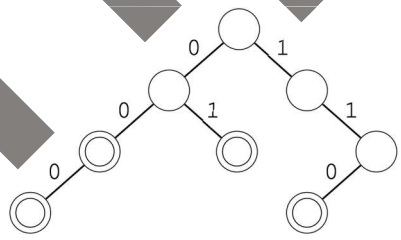



Fig. 3

```
/* Program 2 */
node_t *new_node(void);
int char2bit(char c);

void _insert(node_t *node, char str[], int i) {
    if (str[i] == '\0') {
        [A]
        return;
    }
    int j = char2bit(str[i]);
    if (node->child[j] == NULL) {
        node->child[j] = new_node();
    }
    [B]
}

void insert(node_t *node, char str[]) {
    _insert(node, str, 0);
}
```

```
/* Program 3 */
int exists(node_t *node, char str[]) {
    int i;
    for (i = 0; node != NULL; i++) {
        if (str[i] == '\0') {
            [C]
        } else if (str[i] == '0') {
            [D]
        } else if (str[i] == '1') {
            [E]
        }
    }
    return 0;
}
```

```
/* Program 4 */
int main(void) {
    node_t root = {};
    insert(&root, "01");
    insert(&root, "110");
    return exists(&root, "010");
}
```

Sample

Sample

## Problem 5

I. Answer the following questions about the electron wave functions and quantum states in a one-dimensional system. Suppose that the electron motion is limited in the $x$ direction. $m$ is the electron mass, $\hbar$ is a constant given by the Planck constant $h$ divided by $2\pi$, $k_B$ is the Boltzmann constant, and $i$ is the imaginary unit.

(1) The wave function $\psi(x)$ of a single electron with the eigen energy $E$ follows the time-independent Schrödinger equation

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)\,.$$

Consider the eigen states of an electron inside the potential $V(x)$ shown in Fig. 1 expressed by

$$V(x) = \begin{cases} +\infty & (x \le 0) \\ V_0 & (0 < x < L) \\ +\infty & (x \ge L) \end{cases}.$$

(1-i) Show that the wave function of an electron with the eigen energy $E$ at $0 < x < L$ is expressed by the form of $\psi(x) = A\sin(Bx + \phi)$, where $A$, $B$, and $\phi$ are constants. Express $B$ using $m$, $\hbar$, $V_0$, and $E$.

(1-ii) Find the possible eigen energies $E$ and constants $\phi$ using the boundary conditions.

(1-iii) Write the appropriate expressions which fill in [ I ] and [ II ] in the following sentences.
"The squared modulus of the normalized wave function is called the [ I ] function. From the physics perspective, the integral value of the [ I ] function over $x_1 \le x \le x_2$ means [ II ]."

Consider a system with multiple electrons in $V(x)$. Suppose the eigen energies $E$ do not change from the single-electron system. Electrons are in thermal equilibrium and follow the Fermi-Dirac distribution with the Fermi level $E_F$ at the absolute temperature $T$.

(1-iv) Find the probability that a state with an eigen energy $E$ is occupied by an electron, and find the expectation value of the number of electrons having an eigen energy $E$.

(2) Consider the momentum of an electron in free space. The momentum operator is given by $\hat{p} = -i\hbar(d/dx)$.

(2-i) Consider an electron with the following wave function $\varphi(x)$. Find the expectation value of momentum $\langle\hat{p}\rangle$ of this electron. You may use the relation $\int_0^\infty x^N e^{-\beta x}dx = N!/\beta^{N+1}$, where $N$ is a positive integer and $\beta$ is a complex number with a positive real part.

$$\varphi(x) = \begin{cases} 0 & (x < 0) \\ (2a^{1.5}xe^{-(a+ib)x} & (x \ge 0) \end{cases}$$ ; $a$ and $b$ are positive real numbers.

**(2-ii)** By observing the momentum of the electron, a value $p_1$ is obtained. Describe how the wave function changes by the observation of momentum, and find the wave function right after the observation.

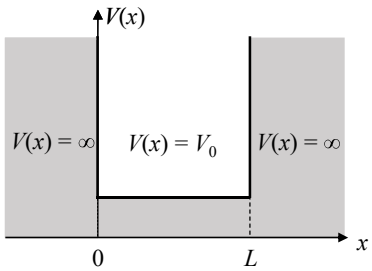



Fig. 1

Sample

II. Consider the metal-oxide-semiconductor (MOS) structure. The energy band diagram at the flat-band condition is shown in Fig. 2. The semiconductor is sufficiently-thick non-degenerate P-type silicon with an impurity concentration of $N_A$. The impurity atoms in the semiconductor are completely ionized. There are no defect states inside the bandgap and at the interface with the oxide. The intrinsic carrier density of silicon is $n_i (\ll N_A)$ and dielectric constant of silicon is $\varepsilon_s$. The oxide has the thickness of $t_{ox}$ and dielectric constant of $\varepsilon_{ox} (< \varepsilon_s)$. A voltage $V_g$ is applied to the metal side with respect to the silicon side. $T$ is the absolute temperature, $\varepsilon_0$ is the vacuum permittivity, $k_B$ is the Boltzmann constant, and $q$ is the elementary charge. Answer the following questions.

(1) At the flat-band condition, express the hole density $p_0$ and electron density $n_0$ in the P-type silicon.

(2) At the flat-band condition, express the Fermi level $E_F$ in the P-type silicon using the intrinsic Fermi level $E_i$.

(3) The silicon will be at the accumulation condition, depletion condition, and inversion condition depending on the applied voltage $V_g$. Draw schematics of the band diagrams of the MOS structure at these three conditions. Also, write the range of voltage $V_g$ for each condition. Indicate the positions of the Fermi level $E_F$ and the intrinsic Fermi level $E_i$ of the silicon in the diagrams. Here, the flat-band voltage and the threshold voltage of the MOS structure are given by $V_{FB}$ and $V_{TH}$, respectively.

(4) Consider the depletion condition using the depletion approximation. When the surface potential of the silicon is $\phi_s$, find the depletion region width $W_D$, area charge density on the metal $\sigma_M$, and the voltage across the oxide $V_{ox}$ as functions of $\phi_s$.

(5) At the depletion condition with the depletion width $W_D$, express the capacitance per unit area $C = d\sigma_M/dV_g$ of the MOS structure in terms of $\varepsilon_0$, $\varepsilon_{ox}$, $\varepsilon_s$, $t_{ox}$, and $W_D$.

(6) Answer whether the following physical quantities become higher or lower as the absolute temperature $T$ decreases. Describe the reasons in 1-2 lines each. Here, assume that the dielectric constants and thicknesses of all layers, impurity ionization rate, and metal work function do not change.

(6-i) Electron density $n_0$ in the P-type silicon at the flat-band condition

(6-ii) Fermi level $E_F$ of the P-type silicon at the flat-band condition

(6-iii) Flat-band voltage $V_{FB}$

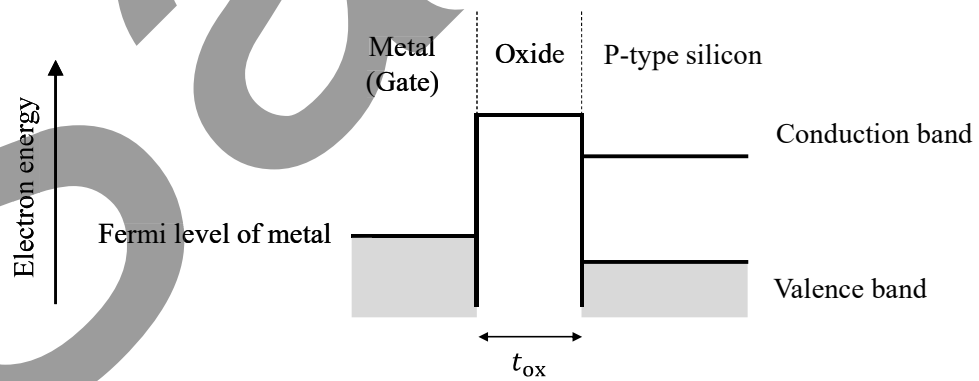



Fig. 2

Sample

## Problem 6

I. There is a plant with the transfer characteristics shown in Fig. 1, where $\tau$ and $J$ are positive constants, and $s$ is a variable of Laplace transforms. A feedback control is applied to this plant using a proportional controller with the gain $K_P$ as shown in Fig. 2. $U(s)$, $Y(s)$, $D(s)$, and $Y^*(s)$ are the Laplace transforms of the input signal $u(t)$, the output signal $y(t)$, the disturbance $d(t)$, and the reference $y^*(t)$, respectively. Answer the following questions.

(1) Find the transfer function $G_0(s) = \frac{Y(s)}{U(s)}$ from the input to the output in Fig. 1.

(2) Find the closed-loop transfer function $G_1(s) = \frac{Y(s)}{Y^*(s)}$ from the reference $Y^*(s)$ of the output $Y(s)$ to $Y(s)$ in Fig. 2.

(3) Find the closed-loop transfer function $G_2(s) = \frac{Y(s)}{D(s)}$ from the disturbance $D(s)$ to the output $Y(s)$ in Fig. 2.

In the following questions, assume $\tau = J = 1.0$.

(4) Sketch a Bode diagram of $G_0(s)$ in Question (1). Using the Bode diagram, explain how $K_P$ affects the stability of the closed-loop system shown in Fig. 2.

(5) Suppose $K_P = \frac{1}{8}$ and $d(t) = 0$ in Fig. 2. Find the output $y(t)$ when the reference $y^*(t)$ is a unit step function. Also sketch a rough shape of $y(t)$.

(6) Suppose $K_P = \frac{1}{8}$ and $y^*(t) = 0$ in Fig. 2. Sketch a rough shape of $y(t)$, and find the final value of $y(t)$ when the disturbance $d(t)$ is a unit step function.

(7) Explain how the waveforms and final values of the unit step responses in Questions (5) and (6) change, especially how the oscillation frequency, decay and stability of the responses are affected, when $K_P$ is increased, using equations and figures if necessary.

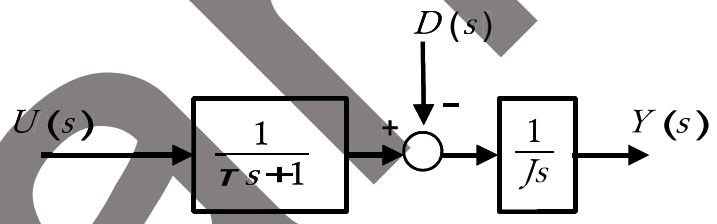



Fig. 1

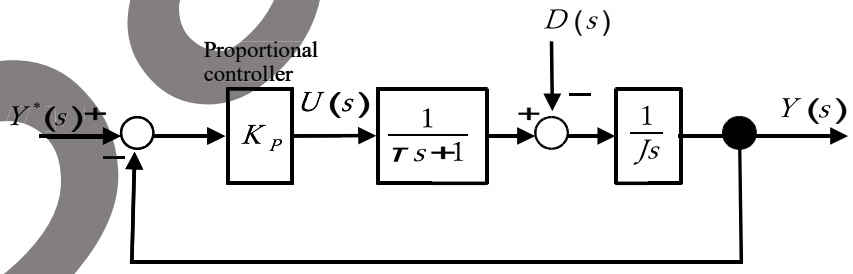



Fig. 2

Sample

II. There is a separately-excited DC machine that uses a permanent magnet as a field magnet. The field flux is constant, and the armature resistance $R_a = 0.5\ \Omega$. For simplicity, the voltage drop between the brushes and the commutator, and the armature reaction are assumed to be negligible. Loss factors other than armature loss, such as iron loss and mechanical loss, are also assumed to be sufficiently small and negligible. Let $\pi$ be circular constant. Answer the following questions.

(1) The DC machine is operated as an electric motor with a power supply of $V_a = 100$ V connected to the armature terminals as shown in Fig. 3. The armature current is $I_a = 20$ A and the armature speed is 1500 rotations per minute. Find the internal electromotive force $E_0$ [V] and the angular speed $\omega_{mot}$ [rad/s] of the rotor. Based on these, find the motor power $P_{out}$ [W] and the motor torque $T_{mot}$ [Nm]. Then find the motor efficiency $\eta_{mot}$.

(2) We connect a load resistor $R_L = 5.0\ \Omega$ to the armature terminals of the same DC machine used in Question (1) as shown in Fig. 4. We then apply a torque to the armature rotor and rotate it. As a result, we obtain an armature voltage $V_a' = 100$ V. Find the angular speed $\omega_{gen}$ [rad/s] and the generator efficiency $\eta_{gen}$.

(3) Explain briefly an advantage of electric vehicles in terms of energy use, based on the energy conversion and efficiency discussed in Questions (1) and (2).

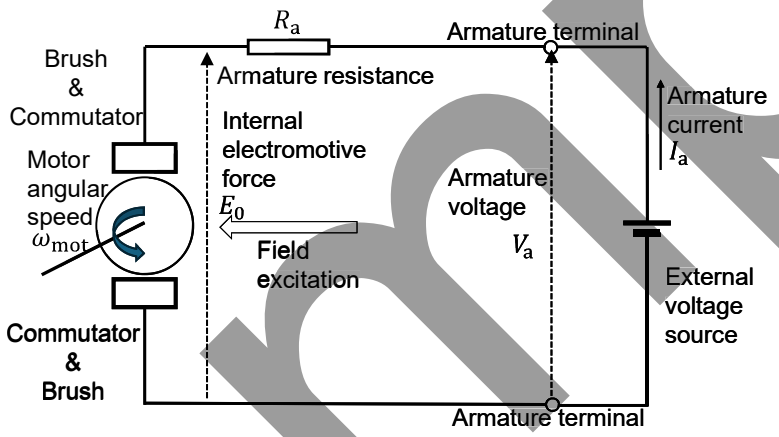



Fig. 3

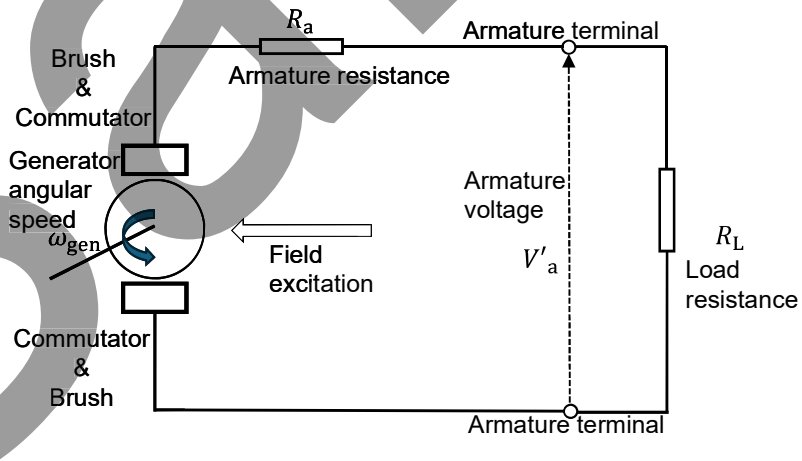



Fig. 4

Sample